

Office Automation Software Lab

Core Course

Official Source Declaration

This lesson is architected using the official **SITTTR Kerala Syllabus (Revision 2021)** for Course Code **1259**. All modules, contact hours, and learning outcomes are mapped directly from the official document.

Course Dashboard

Program	Diploma in Computer Application and Business Management
----------------	---

Course Category	Program Core
Periods per Week	3 (Practical)
Total Periods	45
Prerequisites	Basic Computer Knowledge (Secondary School level)

How to Study this Lab Course

- **Hands-on Practice:** This is a lab course. Every topic mentioned must be practiced on a computer.
- **Step-by-Step Documentation:** Maintain a lab record for every experiment.
- **Business Context:** Always think how these tools apply to a business office (e.g., creating a salary slip in Excel).

Course Outcomes (CO) & Mapping

CO	Description	Hours	Cognitive Level
CO1	Use Word Processing for documents and reports	8	Applying
CO2	Create basic spreadsheets for business	10	Applying
CO3	Apply Advanced Excel formulas and manipulation	12	Applying
CO4	Build PPT presentations and MS Access databases	12	Applying

CO-PO Mapping

Strongly mapped (3) to PO1 (Basic Knowledge) and PO4 (Engineering Tools).

Syllabus Coverage Map

Module	Key Topics	Weightage
1	Word: Mail Merge, Tables, PDF Export	18%
2	Excel: Data Referencing, Logical IF, Auto Sum	22%
3	Advanced Excel: VLOOKUP, Pivot Tables, Dashboards	27%
4	PPT & Access: Business Slides, Database Queries	27%
Exam	Lab Exam I & II	6%

Learning Roadmap

Step 1: Master Document Formatting (Word) → **Step 2:** Learn Data Calculation (Excel Basic) → **Step 3:** Professional Data Analysis (Excel Advanced) → **Step 4:** Data Presentation & Storage (PPT/Access).

Module 1: Word Processing Applications (CO1)

Duration: 8 Hours Focus: Professional Reporting

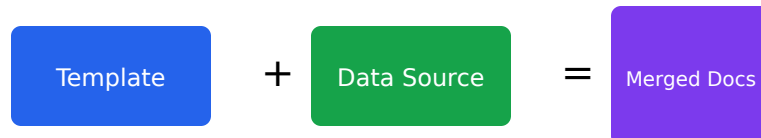
M1.01: Document Creation & Formatting

Creating professional documents involves setting margins, font styles, paragraph spacing, and headers/footers.

Malayalam support: മലയാളം സാക്ഷരതാ നിരക്ക് ഉയർത്തുന്നതിനായി, സർക്കാർ, സ്കൂൾ അധ്യാപകർ, പ്രൈവറ്റ് അധ്യാപകർ, ട്രെയിനിംഗ് സെന്ററുകൾ തുടങ്ങിയവർക്ക് സാക്ഷരതാ പരിപാടികൾ സംഘടിപ്പിക്കാൻ സാധിക്കും.

M1.02: Mail Merge for Multiple Letters

Mail Merge is used to create personalized letters for hundreds of recipients using a single template and a data source (like an Excel list).



The Mail Merge Workflow

M1.03: Tables, Hyperlinks & PDF Export

Module 2: Basic Spreadsheets (CO2)

Duration: 10 Hours Focus: Business Calculations

M2.02: Rows, Columns & Referencing

Understanding **Relative** (A1) vs **Absolute** (\$A\$1) referencing is crucial for copying formulas.

Malayalam support: ുറു ുറുറുറുറു ുറുറുറു ുറുറുറുറുറുറുറു ുറു ുറുറുറുറുറു
ുറുറുറു ുറുറുറുറുറു ുറുറു ുറുറുറുറു (\$) ുറുറുറുറുറുറുറുറുറു. ുറുറുറു Absolute
Reference ുറുറുറു ുറുറുറുറുറുറുറുറു.

M2.03: Basic Formulae & IF Logic

Scenario: Calculate Pass/Fail based on marks (Pass mark = 40).

Formula: =IF(B2>=40, "Pass", "Fail")

Logic: If value in B2 is 40 or more, display "Pass", else "Fail".

Module 3: Advanced Excel (C03)

Duration: 12 Hours Focus: Data Manipulation

M3.01: Charts & Lookup Functions

VLOOKUP: Used to search for a value in the first column of a table and return a value in the same row from another column.

```
=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
```

M3.03: Pivot Tables & Data Protection

M3.04: Basic Dashboard Preparation

Module 4: Presentations & Databases (CO4)

Duration: 12 Hours Focus: Presentation & Storage

M4.01: Professional PowerPoint

- Use **Slide Master** for consistent branding.
- Insert Charts directly from Excel to keep data linked.
- Use **SmartArt** for process flows.

M4.02 & M4.03: MS Access Basics

MS Access is a Relational Database Management System (RDBMS).

Term	Description
Table	Stores data in rows (records) and columns (fields).
Query	A request to retrieve specific data (e.g., List all students from 'Kochi').
Primary Key	A unique ID for each record (e.g., Admission Number).

Official Lab Activities

Activity	Description
Resume Preparation	Create a 2-page professional resume using Word tables and formatting.

Activity	Description
Salary Sheet	Create an automated payroll in Excel using IF, SUM, and HRA/DA percentages.
Sales Dashboard	Pivot table analysis of a yearly sales dataset.
Student DB	Create an Access database for student marks and query the top performers.

Formula & Procedure Bank

Excel: SUMIF

=SUMIF(Range, Criteria, Sum_Range)

Use: Sum values based on a condition.

Excel: CONCATENATE

=A2 & " " & B2

Use: Join First Name and Last Name.

Visual Revision Library

Excel Cell Anatomy

[Column Letter][Row Number] → B15

Access Relationship

Assessment Methodology

Type	Details	Weight
Lab Exam I	Covers Module 1 & 2 (Word & Basic Excel)	1.5 Hours
Lab Exam II	Covers Module 3 & 4 (Adv Excel, PPT, Access)	1.5 Hours
Open Ended	Creative project combining multiple tools	3 Hours

Module-wise Q&A

► Q1: What is the main advantage of Mail Merge? (CO1)

► Q2: Explain the difference between COUNT and COUNTA in Excel. (CO2)

► Q3: What is a 'Slicer' in an Excel Dashboard? (CO3)

► Q4: What is a Primary Key in MS Access? (CO4)

Practice Model Paper

(Not an official examination paper)

Part A (Word - 10 Marks): Create a formal meeting notice. Use Mail Merge to address it to 5 different department heads listed in an Excel sheet.

Part B (Excel - 20 Marks): Create a table with 'Product Name', 'Price', and 'Quantity'. Calculate 'Total'. Use an IF statement to apply a 10% discount if Total > 5000.

Part C (Access - 10 Marks): Create a table 'Employees'. Fields: EmpID (PK), Name, Salary. Insert 5 records. Write a query to show employees with Salary > 25000.

Quick Revision Checklist

- ☒ Can I perform a Mail Merge?
- ☒ Do I know the difference between \$A1, A\$1, and \$A\$1?
- ☒ Can I create a VLOOKUP without errors?
- ☒ Can I create a Pivot Table and add a Chart to it?
- ☒ Do I know how to set a Primary Key in Access?

Glossary

Ribbon

The tabbed toolbar at the top of Office applications.

Workbook

An Excel file containing one or more worksheets.

Query

A specific search or data retrieval command in a database.

Transition

The visual effect that occurs when moving from one slide to another in PPT.

Official References

- R1: Bittu Kumar, Mastering MS Office, V&S Publishers, 2017.
- R2: Michael Price, Office 2019 in Easy Steps, BPB Publications, 2019.
- Online: support.office.com